

Car Accident Analysis

Course: DS/CMPSC 410

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Team Name: The Crash Crew

1. Introduction and Motivation

- Road accidents remain one of the leading causes of injuries and fatalities in the United States, placing heavy burdens on public safety, healthcare, and infrastructure systems. With millions of traffic incidents reported nationwide each year, understanding the patterns and contributing factors behind these accidents is crucial for improving road design, urban planning, and traffic management policies
 - Accident prevention and road planning decisions demand large-scale evidence, not just small-sample studies. The dataset contains many facets and information spanning temporal, spatial, environmental and infrastructure information. All this information will not be easily accounted for using traditional data analysis methods. Big Data will allow us to ensure most of the important elements during the time period are considered.
 - Traditional data tools struggle with the sheer size (millions of records) and heterogeneity (structured + semi-structured) of these datasets. It will take an obscene amount of time to sort through and make sense of this collection of data.
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2. Project Objectives

Clearly state the goals of the project. Example objectives:

- Collection and preprocessing of large-scale data from specific sources.
 - Acquire a large dataset of road accidents from publicly available sources. We will also standardize formats for attributes that involve dates, time, location, and possible casualties.
- Developing PySpark workflows for data cleaning, transformation, and feature extraction.
 - Filter and unduplicate any records we acquire.
 - Use dataframe operations for scalable transformations.

- Implementing scalable analysis (e.g., clustering, classification, regression, time-series analysis).
 - We will use clustering to find “hotspots” at similar times and regions of the accidents.
 - Time series analysis for finding trends in different weathers/seasonal conditions.
 - Visualizing insights and evaluating performance with varying numbers of cores.
 - Off of cluster analysis we intend to create maps of regions of where the accidents occur the most.
 - Compare runtime and scalability using different numbers of cores.
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3. Data Description

- **Data Source:** Where the dataset comes from (public repository, synthetic generation, course-provided).
Source: Kaggle
Link: <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>
 - **Data Size:** Approximate number of records and storage size.
Number of records: 7.73 million
Storage Size: 3.06 GB
 - **Data Characteristics:** Structured/unstructured, temporal/spatial, etc.
Format: Tabular
 - **Data Challenges:** Missing values, skewed distribution, high dimensionality, streaming, etc.
Missing values in several columns
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4. Technical Approach

Tools & Frameworks

- **PySpark** will be the primary framework:
 - DataFrames for scalable transformations and queries.
 - Spark SQL for exploratory aggregations.
 - MLlib for clustering models (e.g., k-means, Gaussian Mixture Models).

- **External libraries (light use):**
 - Matplotlib/Seaborn for visualization after exporting aggregated results.
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Cluster Usage Plan

- Run the project on a multi-node Spark cluster (4 Nodes) to handle the dataset's scale. We may use a single node cluster for troubleshooting and prototyping our project file plan.
 - Partition data by geography (state, county) or time (year, month) to balance workloads across nodes. Our early test cases may focus on smaller time periods/ accident location s
 - Compare clustering runtime and quality under different cluster sizes (e.g., 1, 2 or 4 nodes).
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Expected Job Types

- Batch processing: Cleaning and feature engineering of accident data across multiple years.
 - Iterative ML training: Running clustering algorithms (e.g., k-means with varying k) to identify spatial and temporal accident hotspots.
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Storage Requirements and File Formats

- Input: CSV files (original dataset).
 - Intermediate outputs: CSV for cleaned subsets and aggregated statistics.
 - Final results: CSV for cluster assignments, PNG for plots/heatmaps.
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Pipeline Overview

1. Data Ingestion & Preprocessing

- Load accident data from CSV into Spark DataFrames.
- Handle missing values (e.g., drop or impute weather fields).
- Create features for clustering:
 - Geospatial: latitude, longitude, state.
 - Temporal: hour, weekday, season.
 - Environmental: precipitation, visibility, lighting.

2. Exploratory Analysis (Spark SQL + PySpark)

- Accident counts by state, severity, and year.
- Early heatmaps showing accident density by location.

3. Clustering (MLlib)

- Apply **k-means** (and possibly other clustering algorithms) to group accidents by geography and conditions.
- Evaluate clustering quality with metrics like WSSSE (Within-Set Sum of Squared Errors).
- Run clustering at different scales (1–4 cluster nodes) to test scalability.

4. Visualization & Interpretation

- Export clustered results to CSV for visualization.
- Create geospatial hotspot maps highlighting clusters.
- Interpret clusters to inform road planning insights (e.g., dangerous intersections, poor-weather hotspots).

5. Expected Outcomes

- Describe the anticipated deliverables, such as:
 - Clean, documented PySpark code.

- Scalable analytics or ML models.
 - Visualizations and summary dashboards.
 - Final report and/or presentation.
- Classification accuracy, F1 score, runtime scalability.
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6. Work Plan & Timeline

Week	Goals / Tasks	Deliverable(s)
Week 1 (Project kickoff)	Finalize scope, confirm dataset, create repo, assign roles, schedule weekly meetings	Project charter, repo skeleton, communication plan
Week 2 (Data acquisition)	Download Kaggle dataset, verify sizes, sample small subset, initial schema inspection	Raw data checksum, sample CSVs, schema doc
Week 3 (Preprocessing prototype)	Prototype CSV ingestion into Spark, basic cleaning (drop duplicates, parse timestamps)	PySpark ingestion notebook, cleaning script v0
Week 4 (Feature engineering)	Create temporal and geospatial features, handle missing weather fields, partitioning strategy	Feature spec, cleaned Parquet/CSV subsets
Week 5 (Exploratory analysis)	Aggregations: counts by state/year/hour, early	EDA notebook, initial plots (PNG)

heatmaps, severity
distributions

Week 6 (Clustering experiments)	Run k-means and GMM on sample data, evaluate WSSSE, tune k	Clustering notebooks, evaluation report
Week 7 (Scale-up & cluster tests)	Run clustering across cluster sizes (1,2,4 nodes), measure runtime & memory scaling	Scalability report, runtime graphs
Week 8 (Visualization & mapping)	Export cluster assignments, produce geospatial hotspot maps, finalize dashboard visuals	Maps, PNG heatmaps, visualization notebook
Week 9 (Results interpretation)	Interpret cluster results, tie to road safety insights, prepare writeups	Draft final report sections, slide deck draft
Week 10 (Finalize & Present)	Polish report and slides, rehearse presentation, create README and usage instructions	Final report PDF, slides, code release, presentation

7. Division of Labor

Assign roles among team members, e.g.:

- Data acquisition & preprocessing lead: Rujhan Najib
- PySpark pipeline & ML lead. Joshua George, Minh Khoi Duong

- Visualization & results lead: Jonathan Park
 - Documentation & presentation lead: Aqil Harith Ramsaid
 - All of us will also consistently and regularly check and comment on other people's tasks to ensure a systemic and consistent result.
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8. Potential Challenges & Mitigation

Identify possible risks (data too large, PySpark API learning curve, cluster resource limits) and how the team will address them (data sampling, fallback tools, scheduling batch jobs).

We may have a very large dataset that can exceed our machines and/or cluster memory causing it to slow down. We will combat this by data sampling during the development with something more manageable.

We may also run into quality issues with the data. We will mitigate this by defining "rules" for handling missing or "bad" data.

9. References

List any datasets, papers, or tutorials you are drawing inspiration from.

- <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>
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